

Chapter 6 – Nmap (Network Scanning & Enumeration)

Introduction to Nmap

What is Nmap?

Nmap stands for:

Network Mapper

It is one of the most powerful and widely used tools in:

- Ethical Hacking
 - Cybersecurity
 - Penetration Testing
 - Network Administration
-

What Does Nmap Do?

Nmap scans networks and tells:

- ✓ Which devices are connected
 - ✓ Which systems are online
 - ✓ Which ports are open
 - ✓ Which services are running
 - ✓ Which operating system is used
 - ✓ Service versions
 - ✓ Firewall filtering status
-

Daily Life Example

Imagine a security guard checking:

- Which doors are open

- Which rooms are occupied
- Which guards are active

Similarly, Nmap checks:

- Open ports
- Active systems
- Running services

How Nmap Works?

Nmap works by sending specially crafted packets to target systems.

Then it analyzes responses from the target.

Example Scenario

Suppose:

- Kali Linux = Attacker/Scanner Machine
- Windows Machine = Target

Kali sends packets to Windows.

Windows replies depending on:

- Open ports
- Closed ports
- Firewall rules

Ping & Host Discovery

What is Ping?

Ping checks whether a system is:

Online or Offline

using ICMP packets.

Example

ping 192.168.1.10

What Happens Internally?

- 1 Kali sends ICMP Echo Request
 - 2 Windows replies with Echo Reply
 - 3 Scanner knows host is alive
-

TTL (Time To Live)

What is TTL?

TTL = Time To Live

It limits packet lifetime on the network.

Every router decreases TTL by 1.

Why TTL is Important?

TTL values help identify operating systems.

Common TTL Values

TTL Value Possible OS Family

128	Windows
64	Linux/Unix
255	Network Devices

Example

If ping reply shows:

TTL=128

It usually indicates:

 **Windows System**

If:

TTL=64

It usually indicates:

 **Linux/Unix System**

Examples:

- Kali Linux
 - Ubuntu
 - Red Hat
 - CentOS
-

Port Scanning Concept

What is a Port?

Ports are logical communication endpoints.

Applications/services run on ports.

Common Ports

Port Service

80 HTTP

443 HTTPS

22 SSH

21 FTP

Port Service

3389 Remote Desktop (RDP)



TCP Three-Way Handshake

Before communication starts:

TCP performs handshake.



Steps

- 1 SYN → Request connection
 - 2 SYN-ACK → Accept connection
 - 3 ACK → Final confirmation
-



Nmap SYN Scan Example

Suppose Kali scans:

Port 80



Process

- 1 Kali sends SYN packet
 - 2 Windows replies SYN-ACK if port is open
 - 3 Nmap understands port is open
 - 4 Nmap terminates connection without full handshake
-



Why This is Important?

This makes scanning:

- ✓ Faster
 - ✓ Stealthier
-

✖ Closed Port Behavior

Suppose Kali scans:

Port 3389

and target replies:

RST

or gives no reply.

Then:

✖ Port is closed

✖ Firewall may block packets

📘 RST Meaning

RST = Reset Connection

It means:

🚫 Connection Rejected

🛡 Filtered State

Sometimes Nmap shows:

filtered

📘 Meaning

Nmap cannot determine port state because:

🔥 Firewall is blocking packets.

🧠 Daily Life Example

Imagine:

You knock on a door.

- If someone replies → Open
 - If someone says "go away" → Closed
 - If nobody responds → Filtered/Blocked
-

Nmap Lab Setup

For practice:

Use:

- Kali Linux VM
- Windows VM

Both should be connected to:

 **Same Network**

Example:

192.168.1.x

1. Basic Nmap Scan

 **Syntax**

nmap target_ip

 **Example**

nmap 192.168.1.10

 **What It Does?**

Scans top 1000 common TCP ports.

2. Nmap Host Discovery (Ping Scan)

Purpose

Find live systems without scanning ports.

Syntax

```
nmap -sn network/subnet
```

Example

```
nmap -sn 192.168.1.0/24
```

Meaning of /24

CIDR notation.

Represents:

255.255.255.0

What It Does?

Scans entire subnet for:

Live Hosts

without port scanning.

3. Default Port Scan

Syntax

```
nmap 192.168.1.1
```

What It Does?

Scans:

Top 1000 Common TCP Ports

4. Full Port Scan

Purpose

Scan all 65,535 TCP ports.

Syntax

`nmap -p- 192.168.1.1`

Meaning

-p-

means:

Scan All Ports

Important Note

This scan is:

 Slower

 More detailed

5. Scan Specific Ports

Syntax

`nmap -p 22,80,443 192.168.1.1`

What It Does?

Scans only:

- Port 22
 - Port 80
 - Port 443
-

Filtered State Example

If output shows:

filtered

then:

 Firewall is blocking scan traffic.

6. Service Version Detection

Purpose

Detects service software versions.

Syntax

```
nmap -sV 192.168.1.1
```

Example Output

May show:

Apache 2.4.49

OpenSSH 7.6

Why Version Detection is Important?

Older versions may contain:

Vulnerabilities

Daily Life Example

Old Android versions may have security flaws.

Similarly:

Old server software may be vulnerable.

7. OS Detection

Purpose

Detect target operating system.

Syntax

`nmap -O 192.168.1.1`

What It Uses?

OS fingerprinting based on:

- TCP/IP behavior
 - Packet responses
 - TTL values
-

Important Note

May require:

Root Privileges

8. Aggressive Scan

Purpose

Performs detailed scanning.

Syntax

```
nmap -A 192.168.1.1
```

Includes

Feature	Option
OS Detection	-O
Version Detection	-sV
Script Scan	-sC
Traceroute	Included

Important Warning

Aggressive scan is:

-  Very detailed
 -  Easily detectable
-

9. Scan Multiple Hosts

Syntax

```
nmap 192.168.1.1 192.168.1.2
```

Scan Entire Subnet

```
nmap 192.168.2.0/24
```

10. Scan Targets from File

Syntax

`nmap -iL targets.txt`

Meaning

`-iL`

means:

Input List

11. Stealth Scan (SYN Scan)

Purpose

Performs half-open stealth scanning.

Syntax

`nmap -sS 192.168.1.1`

Meaning

`-sS`

means:

SYN Scan

Advantages

-  Fast
-  Stealthy
-  May bypass some firewalls/IDS

12. UDP Scan

Purpose

Scans UDP services.

Syntax

```
nmap -sU -p 53,67,161 192.168.1.1
```

Common UDP Services

Port Service

53 DNS

67 DHCP

161 SNMP

Important Note

UDP scanning is:

 Slower than TCP

because UDP often gives no response.

13. Timing Templates

Purpose

Controls scan speed.

Syntax

```
nmap -T4 192.168.1.1
```

 Timing Levels

Option Meaning

T0 Paranoid

T1 Sneaky

T2 Polite

T3 Normal

T4 Fast

T5 Insane

 Important Concept

Fast Scan Stealth Scan

Quick Harder to detect

Noisy Slower

 14. Fragmentation Scan

 Purpose

Attempts to evade simple firewalls.

 Syntax

`nmap -f 192.168.1.1`

 Meaning

-f

means:

Fragment Packets

Warning

May:

- Trigger alerts
 - Cause issues
 - Be detected by IDS/IPS
-

15. Saving Scan Results

Normal Output

```
nmap -oN result.txt 192.168.1.1
```

XML Output

```
nmap -oX result.xml 192.168.1.1
```

Greppable Output

```
nmap -oG result.gnmap 192.168.1.1
```

Save All Formats

```
nmap -oA scan_results 192.168.1.1
```

16. IPv6 Scanning

Purpose

Scan IPv6 targets.

✓ Syntax

```
nmap -6 -sV fe80::1
```

📌 Pro Tips

🔥 Full Advanced Scan

```
sudo nmap -sS -sV -O -p- -T4 192.168.1.1
```

📖 What This Includes?

Option Purpose

-sS	SYN scan
-sV	Version detection
-O	OS detection
-p-	All ports
-T4	Fast timing

☀️ 17. Scan Top 100 Ports

```
nmap --top-ports 100 192.168.1.1
```

📖 Why Useful?

Faster scanning of most common ports.

🌐 18. ARP Ping Scan (-PR)

📖 What is ARP?

ARP = Address Resolution Protocol

Used inside local networks to map:

 **IP Address → MAC Address**

 **What is -PR?**

-PR performs:

 **ARP Ping Scan**

 **Syntax**

```
sudo nmap -PR 192.168.1.0/24
```

 **How It Works?**

-  Nmap sends ARP requests
 -  Live hosts reply with MAC addresses
 -  Nmap identifies active systems
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 **Why ARP Scan is Powerful?**

-  Very reliable on local networks
 -  Bypasses some firewall restrictions
 -  Faster than ICMP ping
-

 **Important Note**

ARP scanning works only on:

 **Local Networks (LAN)**

It does NOT work across routers or remote internet targets.

Daily Life Example

ARP is like asking:

"Who has this IP address? Tell me your MAC address."

Legal & Ethical Warning

Nmap is a powerful tool.

Always remember:

 Unauthorized scanning is illegal

 Scanning public systems without permission may violate laws

Use Nmap only for:

-  Personal labs
 -  Practice VMs
 -  Authorized penetration testing
 -  Security audits
-

Quick Revision Table

Command Purpose

nmap IP Default scan

nmap -sn Ping scan

nmap -p- Full port scan

nmap -sV Version detection

nmap -O OS detection

nmap -A Aggressive scan

nmap -sS SYN stealth scan

nmap -sU UDP scan